The urban quality of life premium

Gabriel M. Ahlfeldt
Humboldt University, BSoE, LSE, CEPR & CESifo
Fabian Bald
European University Viadrina, BSoE
Duncan H.W. Roth
IAB, IZA@LISER & CESifo
Tobias Seidel
University of Duisburg–Essen & CESifo

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Motivation

- **Productivity advantages** are the established explanation for cities
  - Abundant evidence for an urban productivity premium (Combes and Gobillon, 2015)
- Intuitively, **cities could also be good places to live**
  - Yet limited evidence for an urban QoL premium (Albouy, 2011)
- This may reflect **measurement error in the Rosen–Roback framework**
  - **Idiosyncratic tastes and local ties** are important (Diamond, 2016)
  - Tradable-goods and local-service prices may matter, too (Diamond & Moretti, 2021)
- Accurate estimation of the **urban QoL premium** is crucial for
  - Understanding origins of widening spatial inequalities
  - Optimal provision of public goods and spatial transfers

What we do

- **Quantify bias in urban QoL premium when ignoring spatial frictions**
  - **Mobility frictions** (idiosyncratic preferences; local ties)
  - **Trade frictions** (gravity; local non-tradable services)
- Use **QSM as data-generating process** in a Monte Carlo study
  - **Specification error** (wrong parameter values ⇒ wrong account of spatial fractions)
  - **Aggregation error** (wrong spatial units don't correspond to preferences ⇒ MAUP)
- **Application for Germany**
  - QoL ranking of locations & urban QoL premium
- Accessible **ABRSQOL-toolkit** with user-friendly syntax available on GitHub
  - Implements numerical solver in `MATLAB, R, Python, Stata`

What we find

- **Monte Carlo:**
  - **Omitting idiosyncratic tastes** $\Rightarrow$ **downward bias** in urban QoL premium
  - **Omitting local ties** $\Rightarrow$ **upward biases** in urban QoL premium
  - Bias from trade frictions depends on productivity, wage and floor space gradients
  - **MAUP** is a real problem, but **not mechanical**: Bias depends on aggregation error
  - **Use model with frictions and economically meaningful spatial units**
- **Application to Germany:**
  - Different QoL rankings (average change in rank: 17)
  - **Urban QoL premium is large and increasing over time**
    - Our estimate: 0.29 (elasticity)
    - Rosen-Roback: 0.18
    - Redding (2016): 0.52
    - Urban wage premium: 0.06 (0.03 conditional on worker fixed effects)
  - **No MAUP**: Same premium aross 400 counties, 257 LLMs, 141 LLMs

Literature & contribution

The model

A QSM with canonical building blocks

- J **locations (local labour markets)**
  - $\bar{L}_m^b$ workers grew up in hometown $m \in J$ (exogenous)
  - $\bar{L} = \sum_m \bar{L}_m^b$ workers in the economy choose residence $i \in J$ (endogenous)
- **Utility** from tradables, non-tradables (services), floor space and QoL
  - **Tradables:** produced from labour
  - **Non-tradables (services):** produced from labour and floor space
  - **Floor space:** produced from capital and land
  - **QoL:** Locational fundamental (akin to TFP in production function)
- **Spatial frictions**
  - **Mobility**: Idiosyncratic taste shocks & local ties (dislike residence $\neq$ hometown)
  - **Trade**: Costly traded goods & non-tradable services
  - **No commuting** (workers live and work in same local labour market)

QSM Stylized facts

Exploit unique mapping for model solution and model inversion

Forward mapping: primitives determine equilibrium outcomes

Primitives
Fundamentals $(A_i, \bar{\varphi}_i, \eta_i, \bar{L}_i^b, \tau_{ij})$ & structural parameters

numerical solution $\longrightarrow$

Endogenous outcomes
population, wages, housing prices, tradable-goods and local-service prices, trade and migration flows

- MC DGP, counterfactuals & generating missing variables (e.g. price indices).

Reverse mapping: observed outcomes reveal fundamentals

Observables
wages, housing prices, population, hometown population, trade and migration

inversion $\longrightarrow$

Fundamentals
including quality of life, A_i

- Exploit this inversion to **measure QoL** and estimate the **urban QoL premium**.

Monte Carlo study

Why a Monte Carlo?

- **Problem:** observational data cannot reveal QoL measurement error
  - True QoL is latent, so there is no observed benchmark for a recovered QoL measure
  - The true spatial units over which households choose locations are also unknown
- **Solution:** use the full QSM as a known data-generating process (DGP)
  - **Draw city primitives**—including QoL—and solve for equilibrium outcomes
  - **Apply QoL inversion** to equilibrium outcomes to obtain QoL measurement
  - **Compare inverted QoL to *ground truth* QoL** known in the Monte Carlo
    - Reveals measurement error and bias in estimated urban QoL premium
- This isolates two measurement problems
  - **Specification error:** incorrect assumptions about spatial frictions
  - **Aggregation error:** observed units differ from true location-choice alternatives

Monte Carlo setting

- **Geography:**
  - $R = 250$ synthetic countries, each with $J = 144$ cities on a 12×12 grid
  - A $500\,\mathrm{km} \times 500\,\mathrm{km}$ area $\Rightarrow$ Approximate Germany and its 141 local labour markets
- **Primitives:**
  - Draw QoL, housing productivity, labour productivity, and hometown population
  - Known *ground truth* parameter values, canonical values or own estimates
    - e.g. idiosyncratic tastes $\gamma = 3$; local ties $\xi = 5.6$ from own estimation
- **Equilibrium:**
  - Solve for populations, wages, prices, trade, and migration for all countries Validation
  - Assume that econometrician observes these equilibrium outcomes
- **Measurement:**
  - Use wrong parameter values or spatial units and invert QoL
  - Compute measurement error (by location) & urban QoL premium bias (by country)

QoL recovery and specification error

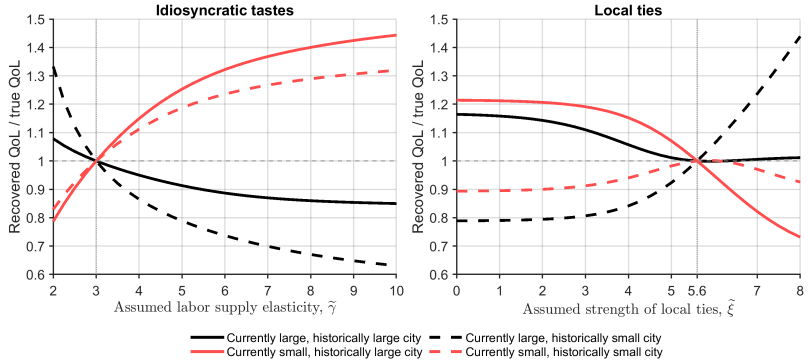
$$\widehat{A}_i = \frac{(\widehat{P}_i^t)^{\alpha\beta}(\widehat{p}_i^n)^{\alpha(1-\beta)}}{\widehat{w}_i}(\widehat{p}_i^H)^{1-\alpha}\left(\frac{\widehat{L}_i}{\widehat{\mathcal{L}}_i}\right)^{1/\gamma} \tag{17}$$

$$\Downarrow$$

$$\begin{aligned}\varepsilon_i = &\underbrace{\left[\widetilde{(1-\alpha)} - (1-\alpha)\right]\ln \widehat{p}_i^H}_{\text{Rosen–Roback}} + \underbrace{\left[\widetilde{(\alpha\beta)} - \alpha\beta\right]\ln \widehat{P}_i^t}_{\text{tradable goods}} \\ &+ \underbrace{\left[\alpha\widetilde{(1-\beta)} - \alpha(1-\beta)\right]\ln \widehat{p}_i^n}_{\text{local services}} + \underbrace{\left(\frac{1}{\widetilde{\gamma}} - \frac{1}{\gamma}\right)\ln \widehat{L}_i}_{\text{idiosyncratic tastes}} \\ &- \underbrace{\left[\frac{1}{\widetilde{\gamma}}\ln \widetilde{\widehat{\mathcal{L}}}_i(\widetilde{\gamma},\widetilde{\xi}) - \frac{1}{\gamma}\ln \widehat{\mathcal{L}}_i(\gamma,\xi)\right]}_{\text{local ties}}\end{aligned} \tag{18}$$

- Hats: **relative difference** to numeraire location
- Tildes: Values assumed or recovered by econometrician: $\varepsilon_i \equiv \ln \widetilde{\widehat{A}}_i - \ln \widehat{A}_i$
- Four channels; parameters can affect **multiple channels**
- $\mathcal{L}$: effective labour supply, including hometown population More
- Restrictions compare inversion error across **model classes**. Model comparison

Mobility frictions determine what city size reveals



- **Omitting idiosyncratic tastes understates QoL in currently large cities**
  - Miss that LA must be attractive to be so large (black lines)
- **Omitting local ties overstates QoL in historically large cities**
  - Miss that LA (to some extent) is large because it was large historically (solid lines)

From measurement error to urban-premium bias

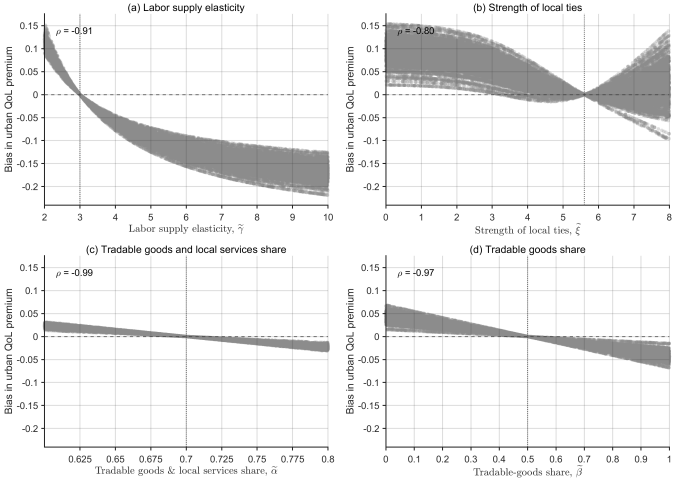
$$B_{rs} \equiv \widetilde{\rho}_{rs} - \rho_r = \frac{\mathrm{Cov}_i\left(\ln \widetilde{\widehat{A}}_{irs} - \ln \widehat{A}_{ir}, \ln L_{ir} \mid r\right)}{\mathrm{Var}_i\left(\ln L_{ir} \mid r\right)}$$

$$\Downarrow$$

$$\begin{aligned} B_{rs} = & \left[\widetilde{(\alpha\beta)}_{rs} - \alpha\beta\right] \frac{\partial \ln P^t_{ir}}{\partial \ln L_{ir}} + \left[\widetilde{\alpha(1-\beta)}_{rs} - \alpha(1-\beta)\right] \frac{\partial \ln p^n_{ir}}{\partial \ln L_{ir}} \\ & - (\widetilde{\alpha}_{rs} - \alpha) \frac{\partial \ln p^H_{ir}}{\partial \ln L_{ir}} + \left(\frac{1}{\widetilde{\widetilde{\gamma}}_{rs}} - \frac{1}{\gamma}\right)\left(1 - \frac{\partial \ln \widehat{\mathcal{L}}_{ir}}{\partial \ln L_{ir}}\right) \\ & - \frac{1}{\widetilde{\widetilde{\gamma}}_{rs}}\left(\frac{\partial \ln \widetilde{\widehat{\mathcal{L}}}_{irs}}{\partial \ln L_{ir}} - \frac{\partial \ln \widehat{\mathcal{L}}_{ir}}{\partial \ln L_{ir}}\right). \end{aligned} \tag{19}$$

- Urban QoL premium is a **descriptive elasticity**
  - $A \uparrow \Rightarrow L \uparrow$
- Mechanical bias from wrong $\tilde{\gamma}$
  - Large city attracts marginal worker with lower taste shock
  - $1/\gamma$ scales the population adjustment $\ln L - \ln \mathcal{L}$
- Bias from wrong $\{\tilde{\alpha}\ \tilde{\beta}, \tilde{\xi}\}$
  - IFF $\{P^t, p^n, p^H, \mathcal{L}\}$ **correlated with city size**

Urban QoL premium: specification bias



- Rosen–Roback limit ($\tilde{\gamma} \to \infty$): **downward**. LA: population reveals marginal-worker QoL.
- No local ties: **upward**. LA: historical size mistaken for QoL.
- Housing or local-services share too low: **downward**—miss WTP revealed by higher prices in large cities.

Modifiable areal unit problem (MAUP)

- **Quantities ($L, \bar{L}^b$) in inversion equation** (unlike in Rosen-Roback)
  - Allow adjustment of idiosyncratic tastes and local ties of marginal workers
- **New challenge**: Find **spatial unit** that corresponds to true choice alternatives
  - Different partitions of spatial data result in different observed $L, \bar{L}^b$ and $\tilde{A}$
  - Wrong aggregations lead to larger $\hat{L}$ and, hence, $\hat{A}$ Illustration
- **Bias in urban QoL** premium from spatial aggregation **theoretically ambiguous**:

$$B_r^{\mathrm{M}} \equiv \widetilde{\rho}_r^{\mathrm{agg}} - \rho_r^{\mathrm{city}} = \frac{\mathrm{Cov}_i\left(\ln \widetilde{A}_{ir}^{\mathrm{agg}}, \ln L_{ir}^{\mathrm{agg}} \mid r\right)}{\mathrm{Var}_i\left(\ln L_{ir}^{\mathrm{agg}} \mid r\right)} - \frac{\mathrm{Cov}_i\left(\ln A_{ir}^{\mathrm{city}}, \ln L_{ir}^{\mathrm{city}} \mid r\right)}{\mathrm{Var}_i\left(\ln L_{ir}^{\mathrm{city}} \mid r\right)}. \quad (20)$$

  - Aggregation can increase artificial correlation of measured $\tilde{A}$ and $L \Rightarrow B^M \uparrow$
  - But can also increase the variance of L across the spatial system $\Rightarrow B^M \downarrow$

Spatial aggregation and the urban QoL premium

- **Monte Carlo procedure**
  - Aggregate true cities into 2×2 'mega cities' Illustration
  - Recover $\tilde{A}$ (true parameters) and compare aggregate ($\tilde{\rho}$) and city-level (ρ) premia

	Random 50%	Largest current L	Largest historical L^b	Highest growth	Highest dispersion	All blocks
Mean MAUP bias	-0.044	-0.042	-0.082	0.012	-0.055	-0.025

- **Headline findings**
  - Aggregation **usually biases the premium downward** (except highest-growth)
  - $\text{Var}(\ln L^{agg}) > \text{Var}(\ln L^{city})$ in every design except „all blocks" Table 2
  - Omitting local ties makes MAUP bias **more positive** in five of six designs No local ties

Monte Carlo takeaways

- Existing QoL inversions are restrictions on our general DGP
  - Rosen-Roback, Allen-Arkolakis (2014), Diamond (2016)
- **Trade-off**
  - Early **Rosen-Roback** models subject to **large specification error, but no MAUP**
  - More **recent frameworks reduce specification error, but subject to MAUP**
- **Monte Carlo insights**
  - Misspecification generates large, directional errors; mobility frictions are central
  - MAUP effect ambiguous and quantitatively smaller

Account for spatial frictions and use economically meaningful units

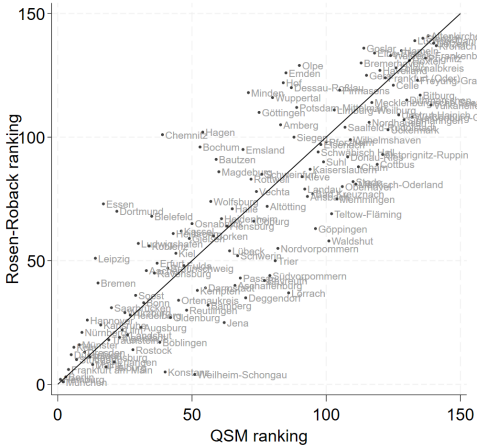
Quantification for Germany

Data (2015) & parameter values

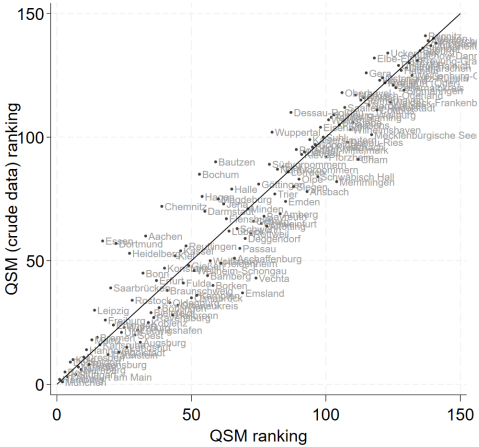
- Price indices
  - Housing prices from Ahlfeldt, Heblich & Seidel (2023)
  - Goods and services prices from Weinand & von Auer (2020)
- Employment and wages from IAB based on universe of German workers
  - Using place of vocational training (Ausbildung) as **hometown**
- $\alpha = 2/3$; from $1 - \alpha = 0.33$ (Federal Statistical Office 2020)
- $\beta = 0.34$ (own estimate)
- $\gamma = 3$ from Krebs and Pflueger (2023)
- $\xi = 5.5$ (own estimate) somewhat larger than for US
  - Consistent with a greater share of workers living in their hometowns

	QSM, best data		Rosen-Roback			QSM, crude data		
	Rank (1)	$\hat{A}$ (2)	Rank (3)	$\hat{A}_{RR}$ (4)	(4)/(2) (5)	Rank (6)	$\hat{A}_{CD}$ (7)	(7)/(2) (8)
Hamburg	1	2.081	2	1.737	0.834	2	1.648	0.792
München	2	2.033	1	1.963	0.965	3	1.644	0.809
Berlin	3	1.851	3	1.678	0.907	1	1.758	0.950
Frankfurt am Main	4	1.696	5	1.520	0.896	4	1.441	0.850
Düsseldorf	5	1.566	12	1.335	0.853	5	1.392	0.889
...	...	...	...	...	...	...	...	...
Freyung-Grafenau	137	0.538	126	0.688	1.278	139	0.547	1.017
Kronach	138	0.526	132	0.651	1.237	140	0.538	1.021
Stendal	139	0.522	141	0.575	1.101	130	0.617	1.183
Vulkaneifel	140	0.519	107	0.773	1.488	141	0.525	1.010
Uelzen	141	0.510	137	0.637	1.248	138	0.548	1.073
Standard deviation		0.276		0.253			0.223	

Full QoL table



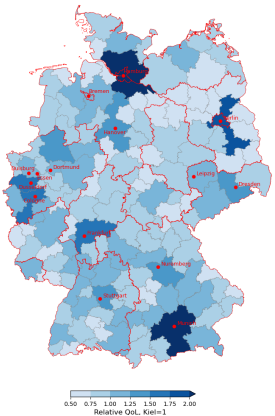
(a) Rosen-Roback (A_{RR})



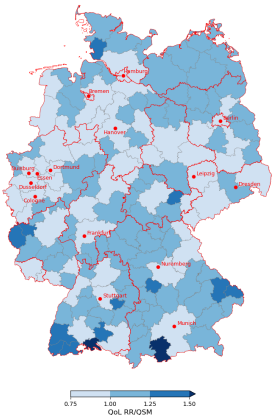
(b) QSM, crude data (A_{CD})

Variation over time

Relative QoL

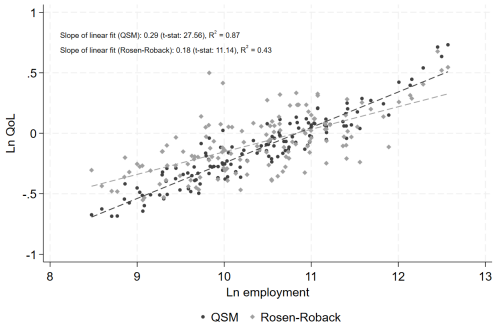


(a) QSM

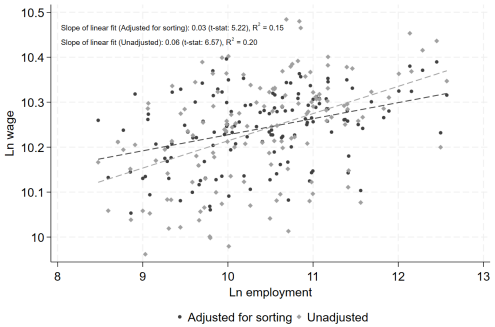


(b) RR/QSM

Urban quality-of-life premium vs. urban wage premium



(a) Urban quality of life premium



(b) Urban wage premium

- Larger urban QoL premium when we account for spatial frictions
- **Urban QoL premium exceeds the urban wage premium**!

Specification-dependency of the estimated urban QoL premium

Model and included spatial frictions	Estimate
Full model	0.298^{***}
All spatial frictions	(0.010)
Rosen–Roback	0.185^{***}
No spatial frictions	(0.017)
AA (no isomorphism)	0.185^{***}
Tradable goods	(0.017)
AA (taste isomorphism)	0.519^{***}
Tradable goods and idiosyncratic tastes	(0.017)
AA (taste isomorphism) + local services	0.521^{***}
All except local ties	(0.017)

Notes: The full model corresponds to Diamond (2016) plus tradable goods. AA denotes Allen–Arkolakis (2014). The taste isomorphism maps Fréchet worker heterogeneity in AA into the idiosyncratic-location-taste correction in our model ($\theta^{AA} = \alpha\gamma$).

- Central estimate: 0.298
  - Doubling city size implies about 22% higher QoL
- Mobility frictions matter
  - No mobility frictions: much lower estimate
  - No local ties: much higher estimate
  - Mirrors the Monte Carlo results
- Trade frictions matter less
  - Depends critically on the measurement of prices

Time and MAUP

Spatial units	2011	2019
400 counties	0.3090***	0.3363***
	(0.0102)	(0.0136)
257 LLMs	0.3091***	0.3448***
	(0.0091)	(0.0122)
141 LLMs	0.3024***	0.3408***
	(0.0124)	(0.0144)

Standard errors in parentheses. $^{***}p < 0.01$. LLMs: local labour markets.

- Preferred specification: full model with all spatial frictions
- Urban QoL premium increases from 2011 to 2019
  - Similar estimates across the three spatial delineations

Conclusions

Take-aways

- **Prioritize economic specification** (address spatial frictions)
  - Bias is larger and directional, whereas MAUP bias is smaller and ambiguous
- **Take spatial units seriously**
  - Align choice unit and mobility parameters, then test alternative delineations
- **Reassess the origins of widening spatial disparities**
  - Urban concentration may reflect QoL—not productivity alone
- **Causal effect** of agglomeration on QoL is a **separate research agenda**
  - Akin to agglomeration effect on productivity

Additional material

Literature (selection) & contributions

- **QoL measurement and public-good valuation (no spatial frictions)**
  - **Theretical framework**: Rosen (1979), Roback (1982)
  - **QoL**: Albouy, Leibovici & Warman (2013); Albouy & Lue (2015); Albouy & Stuart (2020); Blomquist, Berger & Hoehn (1988); Gabriel & Rosenthal (2004), Glaeser (2011), Glaeser & Gottlieb (2009), Shapiro (2006)
  - **Public goods**: Chay and Greenstone (2005) for clean air; Linden and Rockoff (2008) for safety; Cellini, Ferreira & Rothstein (2010) quality of public schools; Greenstone (2017) for review

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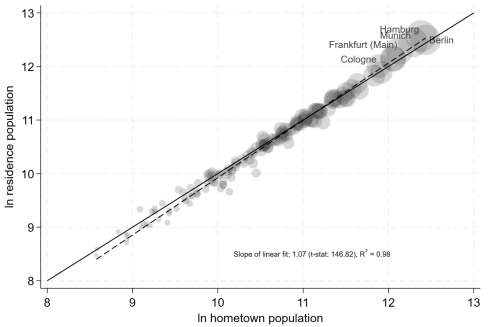
Literature (selection) & contributions

- **Spatial frictions (but no QoL measurement)**
  - **Heterogeneous tastes**: Moretti (2011), Ahlfeldt et al. (2015), Diamond (2016), Monte et al. (2018), etc.
  - **Local ties:** Zabek (2024), etc.
  - **Tradable goods prices**: Armington (1969), Allen & Arkolakis (2014), Redding (2016), Redding and Rossi-Hansberg (2017), etc.
  - **Non-tradable prices**: Moretti & Diamond (2021)

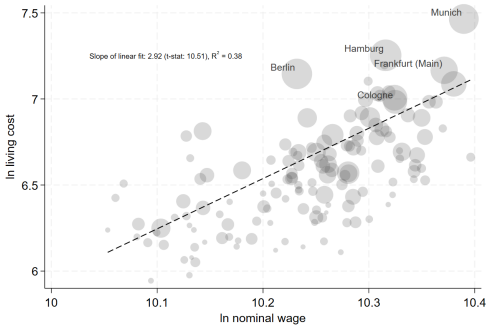
- $\Rightarrow$ We quantify the role of spatial frictions for measurement of QoL
- $\Rightarrow$ We use the quantitative spatial model as data-generating process

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Stylized facts I



(a) Hometown vs. residence population

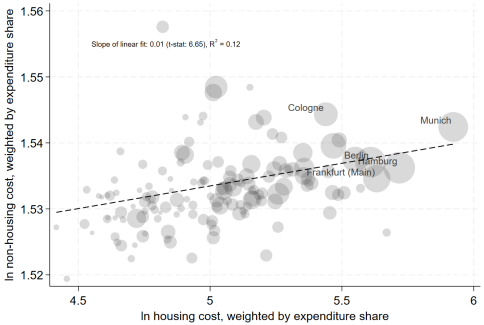


(b) Nominal wage vs. living cost

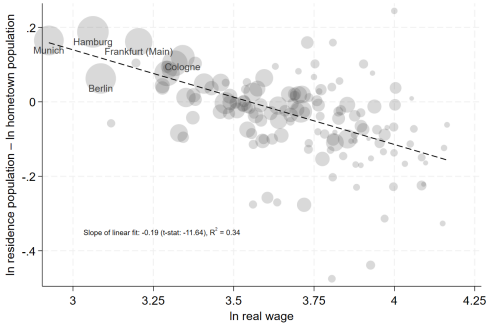
Notes: Unit of observation are the 141 local labour markets (LLM) in Germany.

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Stylized facts II



(c) Housing vs. non-housing cost



(d) Real wage vs. residence pop. surplus

Notes: Unit of observation are the 141 local labour markets (LLM) in Germany.

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Local ties: Ψ_m^b and effective labour supply

$$\widehat{A}_i = \frac{(\widehat{P}_i^t)^{\alpha\beta}(\widehat{p}_i^n)^{\alpha(1-\beta)}}{\widehat{w}_i}(\widehat{p}_i^H)^{1-\alpha}\left(\frac{\widehat{L}_i}{\widehat{\mathcal{L}}_i}\right)^{1/\gamma} \tag{17}$$

$$\mathcal{L}_i = (\exp[\xi]-1)\Psi_i^b \bar{L}_i^b + \sum_{m\in J}\Psi_m^b \bar{L}_m^b,$$

$$\Psi_m^b = \left[1 + \frac{(\exp[\xi]-1)(A_m w_m/\mathcal{P}_m)^\gamma}{\sum_{j\in J}(A_j w_j/\mathcal{P}_j)^\gamma}\right]^{-1}$$

- $\Psi_m^b < 1$: adjustment for workers born in m who leave their hometown
- $\mathcal{L}_i$: effective labour supply to location i
  - Hometown advantage of i
  - National pool of potential movers
- If $\xi = 0$: $\Psi_m^b = 1$ and $\mathcal{L}_i$ is identical across destinations
- A large hometown population can sustain large L_i without high A_i

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Preferences

- Worker ω from hometown m living in city i derives utility from the consumption of goods ($C_{i\omega}$) and floor space ($h_{i\omega}$) according to

$$U_{im\omega} = \left(\frac{C_{i\omega}}{\alpha}\right)^{\alpha} \left(\frac{h_{i\omega}}{1-\alpha}\right)^{1-\alpha} \exp[a_{im\omega}], \tag{1}$$

where $C_{i\omega} = (Q^t_{i\omega}/\beta)^{\beta}(q^n_{i\omega}/(1-\beta))^{1-\beta}$ with

$$Q^t_{i\omega} = \left[\sum_{j\in J} \left(q^t_{ji\omega}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{2}$$

and $q^n_{i\omega}$ (local non-traded services).

Mobility frictions

- The idiosyncratic amenity component $\exp[a_{im\omega}] = \exp[a_{i\omega} + \mathbb{1}\{m = i\} \cdot (\xi/\gamma)]$ is modelled as a stochastic preference shock for each location i, that is shifted upwards if the residence corresponds to the hometown. In particular, $a_{i\omega}$ is drawn from a type-I-extreme value (Gumbel) distribution:

$$F_i(a) = \exp\left(-\tilde{A}_i \exp\{-[\gamma a + \Gamma]\}\right) \quad \text{with} \quad \gamma > 0, \tag{3}$$

  where $\tilde{A}_i \equiv (A_i)^\gamma$ represents the mean of the amenity shock and Γ is the Euler-Mascheroni constant.
- A_i serves as an exogenous measure of quality of life (QoL).
- γ governs the dispersion of individual amenity shocks for each group $\Rightarrow$ introduces imperfect spatial arbitrage $\Rightarrow$ inverse measure of mobility frictions.

Technology: Floor space

- Supplied under perfect competition combining capital (supplied at an exogenous rate r_i^K) with location-specific land, $\bar{T}_i$:

$$H_i^S = \eta_i \left(\frac{\bar{T}_i}{\delta} \right)^{\delta} \left(\frac{K_i}{1-\delta} \right)^{1-\delta}. \tag{4}$$

- Total factor productivity η_i captures the role of regulatory (e.g. height regulations) and physical (e.g. a rugged surface) constraints (Saiz 2010).
- Floor space used for housing and as an input for local services.

Technology: Tradables

- Each location produces a unique variety of a tradable intermediate good (Armington 1969) using labor, L_i^t, as the sole production input under perfect competition according to $q_i^t = \varphi_i L_i^t$
  - Varieties from different origins are combined in a CES consumption aggregate
- Labour productivity $\varphi_i = \bar{\varphi}_i L_i^{\zeta}$ is increasing in local employment according to agglomeration elasticity ζ.
- $\tau_{ji} > 1$ units have to be shipped from j for one unit to arrive in i.
- Perfect competition implies $p_{ji}^t = \tau_{ji} w_j / \varphi_j$

Technology: Local services

- Requires both labour and floor space:

$$q_i^n = \nu_i^n \left(\frac{L_i^n}{\mu}\right)^{\mu} \left(\frac{H_i^n}{1-\mu}\right)^{1-\mu}, \tag{5}$$

  where H_i^n denotes floor space input.
- Perfect competition implies $p_i^n = \nu_i^n (w_i/\varphi_i)^{\mu} (p_i^H)^{1-\mu}$ and floor-space prices p_i^H.

Location choice

- Under the distributional assumptions on the idiosyncratic utility component, we obtain the probability that a worker from hometown m lives in location i:

$$\lambda_{im} = \frac{(A_i w_i / \mathcal{P}_i)^{\gamma} \cdot \exp[\mathbb{1}\{m = i\} \cdot \xi]}{\sum_{j \in J} (A_j w_j / \mathcal{P}_j)^{\gamma} \cdot \exp[\mathbb{1}\{m = j\} \cdot \xi]}, \tag{6}$$

where $\mathcal{P}_i \equiv (P_i^t)^{\alpha\beta} (p_i^n)^{\alpha(1-\beta)} (p_i^H)^{1-\alpha}$ is the aggregate consumer price index.

Location choice

- Summing over all hometown probabilities $\Rightarrow$ residential choice probability:

$$\lambda_i = \sum_m \lambda_{im} = \frac{(A_i w_i / \mathcal{P}_i)^\gamma}{\sum_{j \in J} (A_j w_j / \mathcal{P}_j)^\gamma} \frac{\mathcal{L}_i}{\bar{L}}, \tag{7}$$

where

$$\mathcal{L}_i \equiv (\exp[\xi] - 1)\, \Psi_i^b \bar{L}_i^b + \sum_{m \in J} \Psi_m^b \bar{L}_m^b$$

is the effective labour supply to location i, and

$$\Psi_m^b = \left(1 + \frac{(\exp[\xi] - 1)\,(A_m w_m / \mathcal{P}_m)^\gamma}{\sum_{j \in J} (A_j w_j / \mathcal{P}_j)^\gamma}\right)^{-1} < 1$$

ensures that the residential-choice probabilities of workers from m across all destinations sum to 1

General equilibrium

- Given model primitives, a general equilibrium of the economy is referenced by a vector of the endogenous objects $\mathbf{V} = \{L_i^n, L_i^t, p_i^H, p_i^n, P_i^t, r_i, w_i\}$.
- They are jointly determined by
  - Market clearing for tradables and services
  - Floor-space and land market clearing
  - National labour-market clearing: $L_i = \lambda_i \bar{L}$ with $\sum_{i \in J} \lambda_i = 1$
  - Local labour resource constraint: $L_i = L_i^n + L_i^t$
  - Aggregate consumer price index: $\mathcal{P}_i$

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Empirical and Monte Carlo moments

	Synthetic countries			Data
Outcome	Mean	10th pct.	90th pct.	German cities
Population	0.632	0.581	0.682	0.876
Wages	0.181	0.169	0.193	0.133
Housing prices	0.286	0.261	0.310	0.439
Tradable-goods prices	0.062	0.057	0.067	0.009
Local-service prices	0.074	0.068	0.079	0.032

- Entries are standard deviations of log outcomes across cities.
  - Synthetic columns summarize within-country moments across 250 countries; the data column covers 141 German local labour markets.

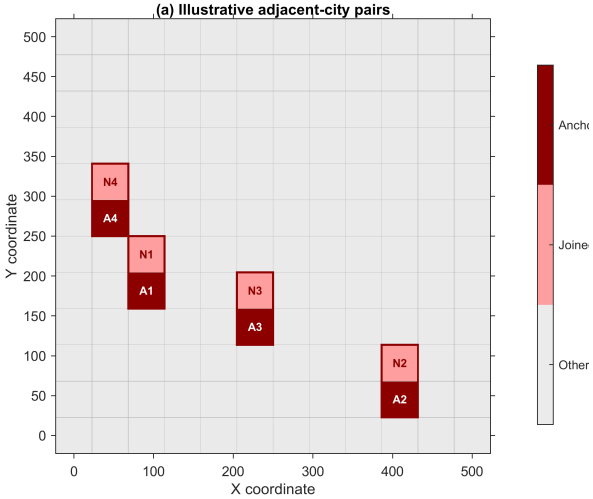
Back to Monte Carlo setting

Table 1: Comparing QoL inversions

Authors	Amenity difference: author measure minus DGP	Omitted spatial frictions
Roback (1982)	$\varepsilon_i^{RR} = -\alpha\beta \ln \widehat{P}_i^t - \alpha(1-\beta) \ln \widehat{p}_i^n - \frac{1}{\gamma}(\ln \widehat{L}_i - \ln \widehat{\mathcal{L}}_i)$	All four
Allen–Arkolakis (2014)	$\varepsilon_i^{AA,H} = \alpha(1-\beta)(\ln \widehat{P}_i^t - \ln \widehat{p}_i^n) - \frac{1}{\gamma}(\ln \widehat{L}_i - \ln \widehat{\mathcal{L}}_i) + (1-\alpha) \ln \widehat{\widehat{\eta}}_i + \delta(1-\alpha) \ln \widehat{\overline{T}}_i$	Local services, idiosyncratic tastes, local ties + housing-supply fundamentals
Allen–Arkolakis (2014), taste isomorphism	$\varepsilon_i^{AA,H+T} = \alpha(1-\beta)(\ln \widehat{P}_i^t - \ln \widehat{p}_i^n) + \frac{1}{\gamma} \ln \widehat{\mathcal{L}}_i + (1-\alpha) \ln \widehat{\widehat{\eta}}_i + \delta(1-\alpha) \ln \widehat{\overline{T}}_i$	Local services, local ties + housing-supply fundamentals
Hsieh–Moretti (2019), imperfect mobility	$\varepsilon_i^{HM} = -\alpha\beta \ln \widehat{P}_i^t - \alpha(1-\beta) \ln \widehat{p}_i^n + \frac{1}{\gamma} \ln \widehat{\mathcal{L}}_i$	Tradable goods, local services, local ties
Redding (2016)	$\varepsilon_i^{R} = \alpha(1-\beta)(\ln \widehat{P}_i^t - \ln \widehat{p}_i^n) + \frac{1}{\gamma} \ln \widehat{\mathcal{L}}_i$	Local services, local ties
Diamond (2016)	$\varepsilon_{jz}^{D} = -\alpha\beta \ln \widehat{P}_j^t$	Tradable goods

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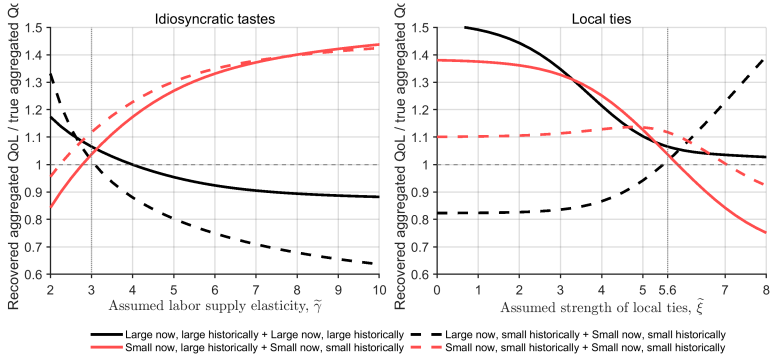
How we illustrate the MAUP



- Four adjacent-city pairs with different current and hometown populations.
- Treat each pair as one observed choice alternative.
  - Sum populations.
  - Population-weight wages, prices, and benchmark QoL.
- Recover QoL, $\tilde{A}$, from aggregated outcomes.

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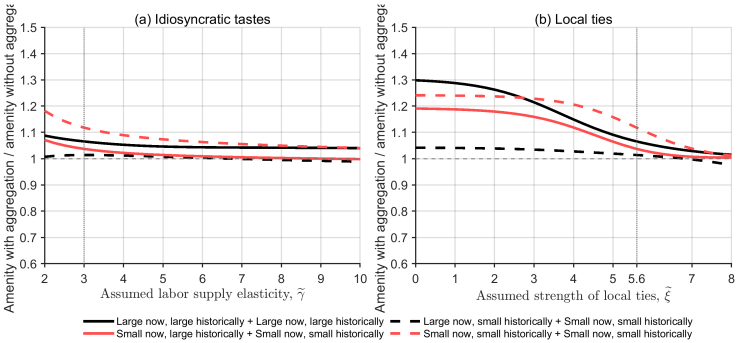
Figure 2b: Total error after aggregation



- Total error combines **parameter misspecification** and **spatial aggregation**.
- The two errors can offset for particular pairs, but the cancellation is accidental.

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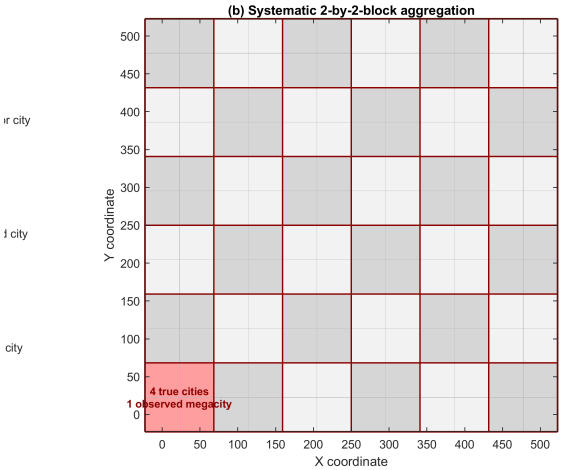
Figure 2c: Pure aggregation error



- Pure MAUP effect (total error - specification error):
  - Compare separate-city recovery under the **same assumed parameters**.
  - At the true parameters, aggregation raises recovered QoL
- Rosen-Roback ($\tilde{\gamma} \to \infty$) insensitive to MAUP as expected

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Systematic aggregation procedure



- Partition the 12×12 economy into 36 non-overlapping 2×2 blocks.
- Merge either all blocks or 18 selected blocks.
  - Randomly or by current population, historical population, growth, or dispersion.
  - Unselected cities remain separate.
- Re-invert QoL and estimate the premium across the observed units.

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Table 2: Covariance–variance decomposition

	Random 50%	Largest current L	Largest historical L^b	Highest growth	Highest dispersion	All blocks
Mean MAUP bias	-0.044	-0.042	-0.082	0.012	-0.055	-0.025
Recovered aggregate premium	0.230	0.231	0.192	0.286	0.219	0.248
$\mathrm{Cov}(\ln \widetilde{A}^{agg}, \ln L^{agg})$	0.150	0.191	0.147	0.192	0.124	0.024
$\mathrm{Var}(\ln L^{agg})$	0.654	0.824	0.762	0.673	0.566	0.095
$\mathrm{Cov}(\ln A^{city}, \ln L^{city})$	0.092	0.092	0.092	0.092	0.092	0.092
$\mathrm{Var}(\ln L^{city})$	0.338	0.338	0.338	0.338	0.338	0.338

- Population variance rises in every selective design, but falls when all blocks are aggregated.
- Whether the premium rises or falls depends on the covariance response relative to this variance response.

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Aggregation without local ties

	Random 50%	Largest current L	Largest historical L^b	Highest growth	Highest dispersion	All blocks
Mean MAUP bias	-0.016	-0.016	-0.011	-0.025	-0.016	0.012
Recovered aggregate premium	0.349	0.349	0.354	0.340	0.349	0.377
$\mathrm{Cov}(\ln \widetilde{A}^{agg}, \ln L^{agg})$	0.228	0.288	0.270	0.229	0.197	0.036
$\mathrm{Var}(\ln L^{agg})$	0.654	0.824	0.762	0.673	0.566	0.095
$\mathrm{Cov}(\ln \widetilde{A}^{city}, \ln L^{city})$	0.123	0.123	0.123	0.123	0.123	0.123
$\mathrm{Var}(\ln L^{city})$	0.338	0.338	0.338	0.338	0.338	0.338

- Without local ties, recovered QoL loads more directly on observed population: QoL–population covariances rise while population variances are unchanged.
- MAUP bias is smaller and closer to zero in most designs, but remains design-dependent.

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